

Eye Inpainting for Group Photos

Group : Team 23

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Outline

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- **Main Approach**
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Introduction

When taking group photos, it's common for a perfectly lit, well-angled, and beautifully composed photo to be ruined by someone with their eyes closed, while other photos may have open eyes but fall short in overall quality. This issue is particularly frequent in large group settings, leading to repeated retakes that are time-consuming and stressful. To address this, we developed an efficient and natural correction technique that intelligently replaces closed eyes, preserving the ideal lighting and composition to create a perfect group photo where everyone looks their best, enhancing the shooting experience and ensuring every photo becomes a cherished memory.



Related Work

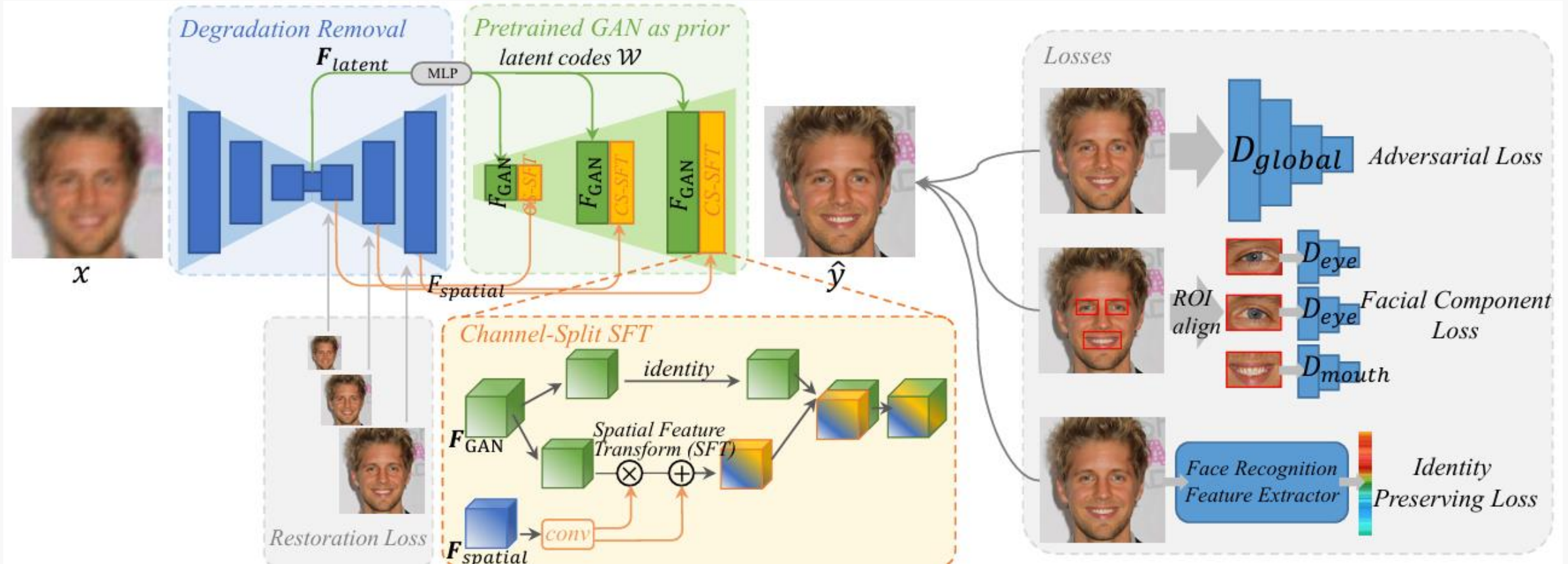
- **Caffe Framework**

A deep learning framework made with expression, speed, and modularity in mind. It is developed by Berkeley AI Research (BAIR) and by community contributors. In this project, Caffe is used to build an SSD(Single Shot Multi-Box Detector) object detection model based on the ResNet-10 architecture for detecting faces in images.

- **Dlib**

A modern C++ toolkit containing machine learning algorithms and tools for creating complex software in C++ to solve real world problems. In this project, Dlib is used to load a face landmark detection model for locating the position of the eyes.

Related Work - GFPGAN



Dataset

Celeb-ID

In this repository, we provide a benchmark eye-inpainting dataset called Celeb-ID. We do not provide the images, but we include scripts to download, align, and process the images, which results in a dataset of over 110,000 images of roughly 17,000 different celebrities



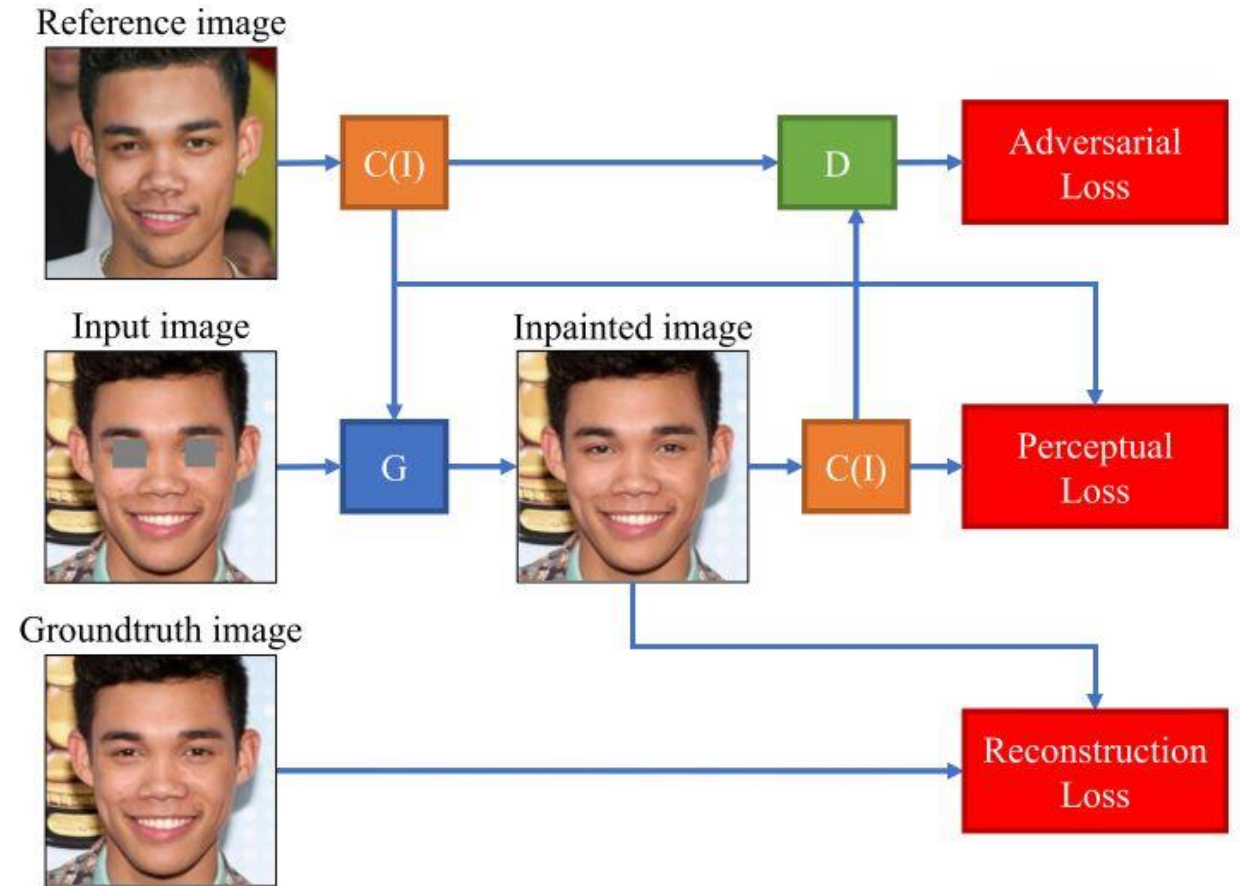
Reference: https://github.com/bdol/exemplar_gans#celeb-id-benchmark-dataset

Platform

Training Environment		Module	
CPU	AMD Ryzen 5 7600X	python	3.9.21
RAM	32 GB	cog	0.13.6
GPU	NVIDIA GeForce RTX 4070	face-recognition	1.3.0
System	Windows 11	basicsr	1.4.2
		realesrgan	0.3.0
		scipy	1.13.1
		dlib	19.24.2
		pytorch	2.4.1
		opencv	4.10.0
		numpy	1.26.4
		torchaudio	2.4.1
		torchvision	0.19.1
		tensorboard-data-server	0.7.2

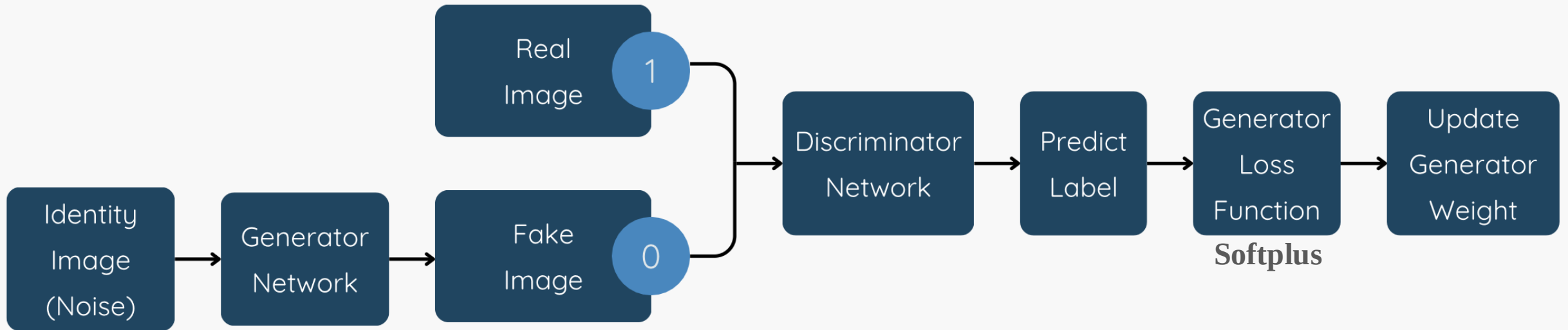
Baseline – ExemplarGAN

ExemplarGANs are a type of conditional GAN that utilize exemplar information to produce high-quality, personalized in-painting results. We propose using exemplar information in the form of a reference image of the region to in-paint, or a perceptual code describing that object. Unlike previous conditional GAN formulations, this extra information can be inserted at multiple points within the adversarial network, thus increasing its descriptive power. We show that ExemplarGANs can produce photo-realistic personalized in-painting results that are both perceptually and semantically plausible by applying them to the task of closed-to-open eye in-painting in natural pictures. A new benchmark dataset is also introduced for the task of eye in-painting for future comparisons.



Baseline – ExemplarGAN (Cont.)

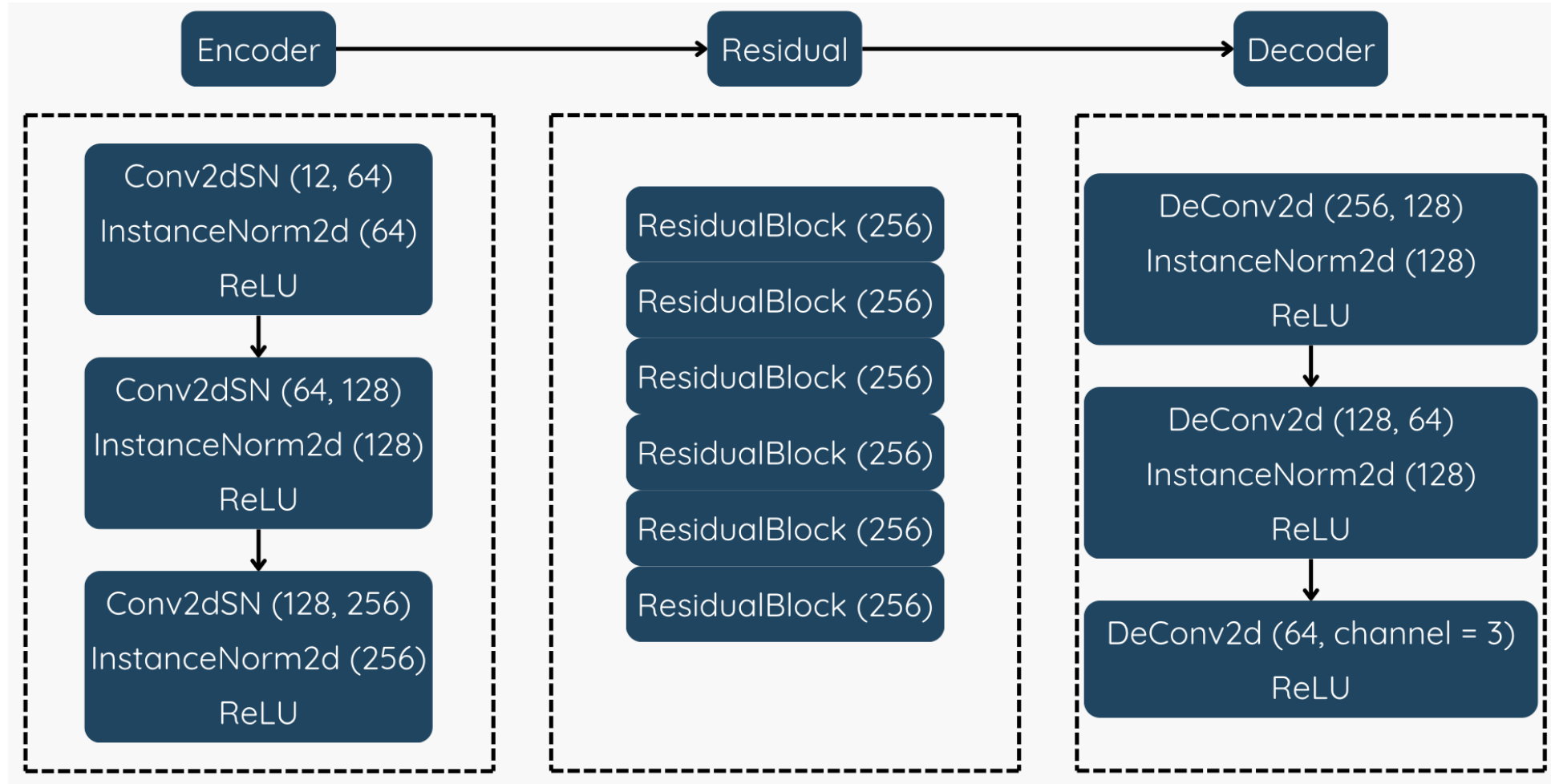
Training Generator



$$\mathcal{L}_D = \frac{1}{N} \sum_{i=1}^N i = \frac{1}{N} [\log(1 + e^{-d_{Fake}^{(i)}})]$$

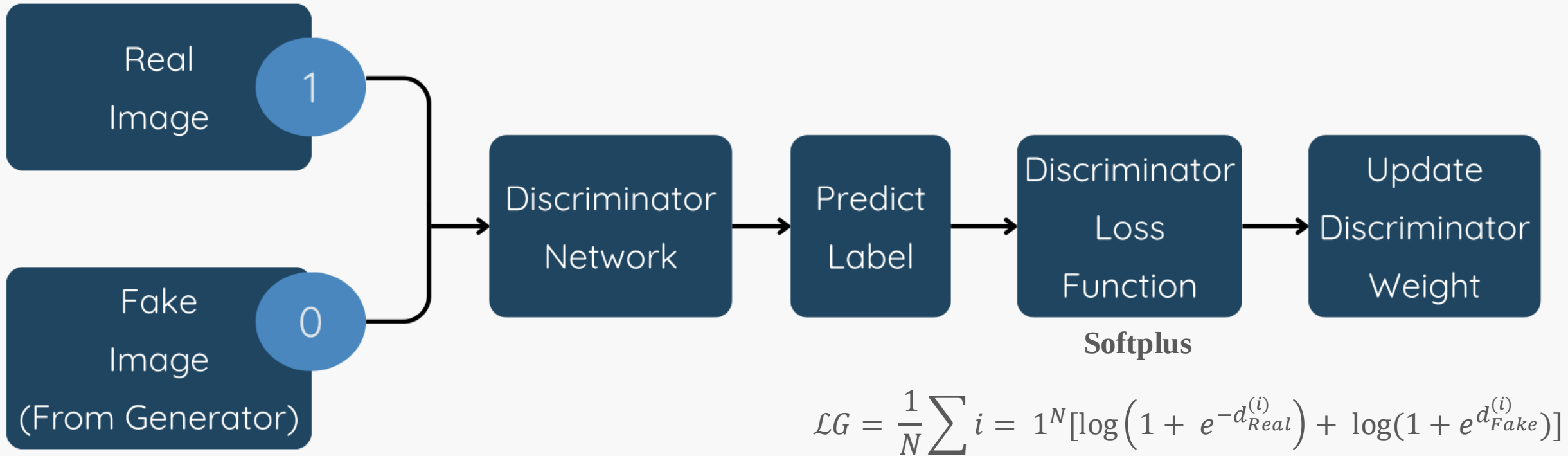
Baseline – ExemplarGAN (Cont.)

Generator Encoder-Residual-Decoder



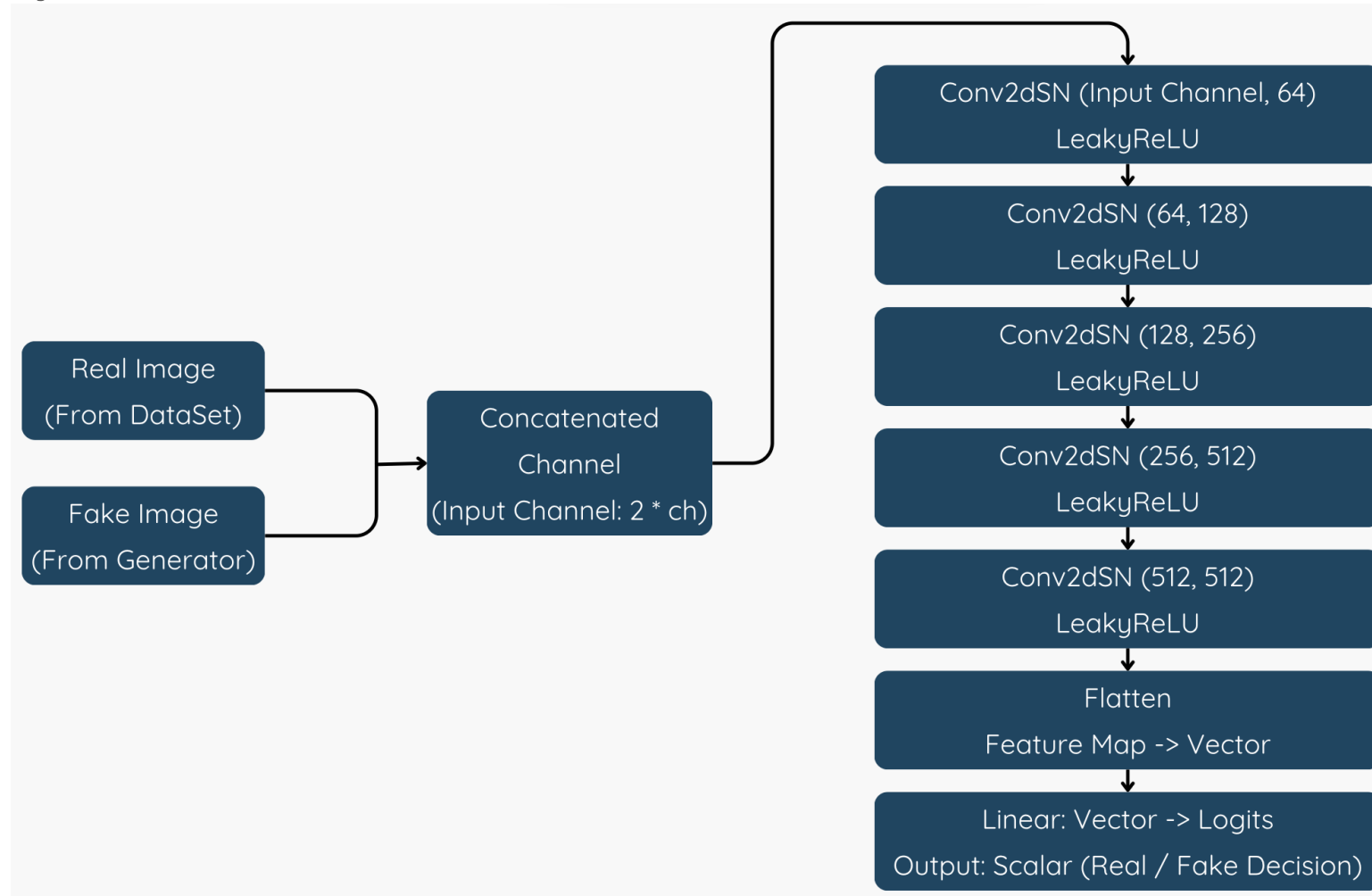
Baseline – ExemplarGAN (Cont.)

Training Discriminator



Baseline – ExemplarGAN (Cont.)

Discriminator Layer



Baseline

Why Choose ExemplarGAN as our baseline?

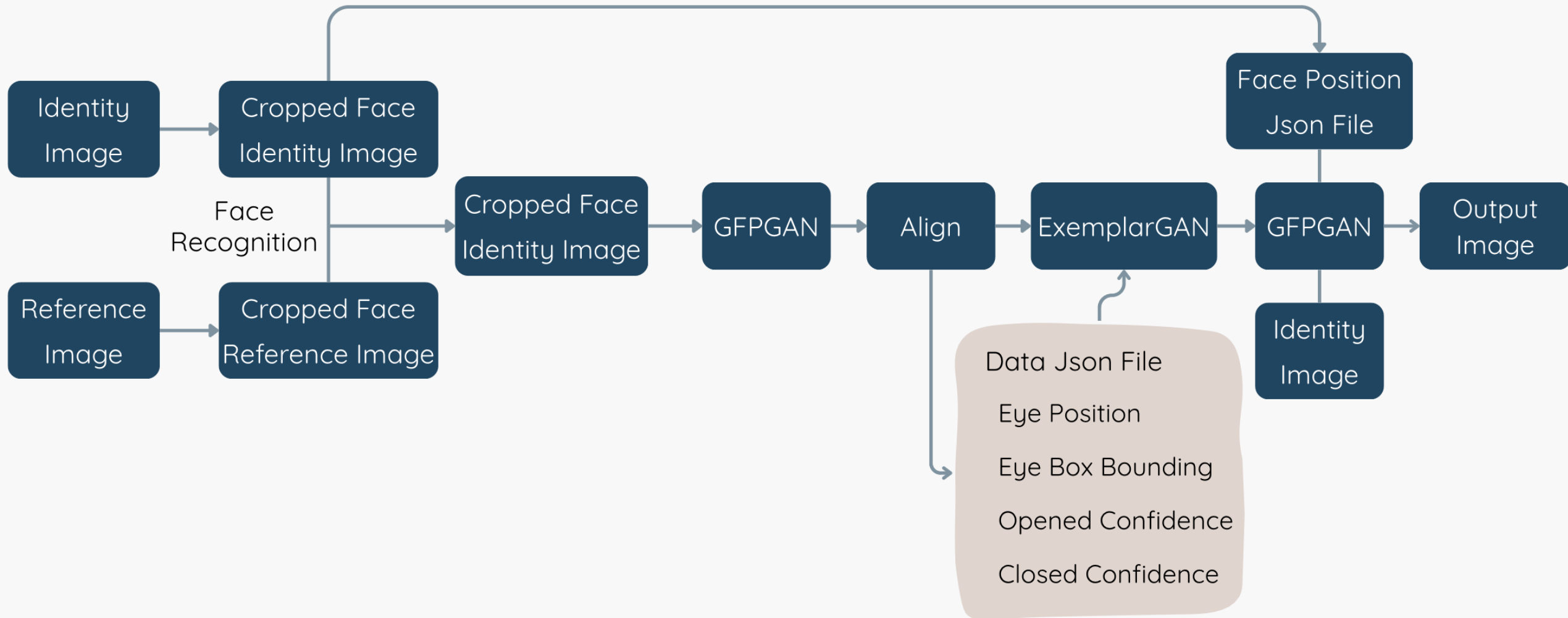
- It performs exceptionally well in single-person eye restoration.
- Using the reference image to guide the generation process ensures that the generated eyes align with the original person.



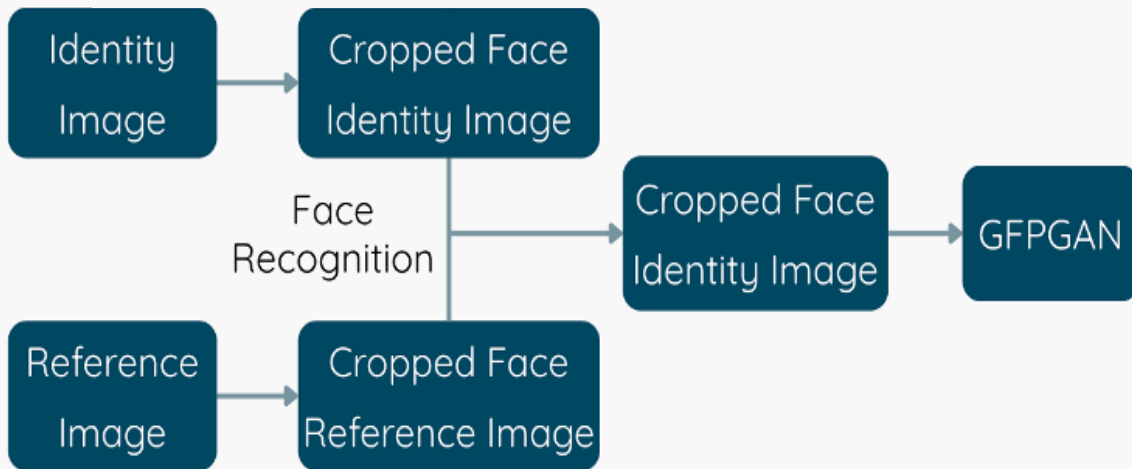
What did we achieve using this baseline?

- We extended the eye restoration technology to handle multi-person group photos.

Main Approach – Architecture

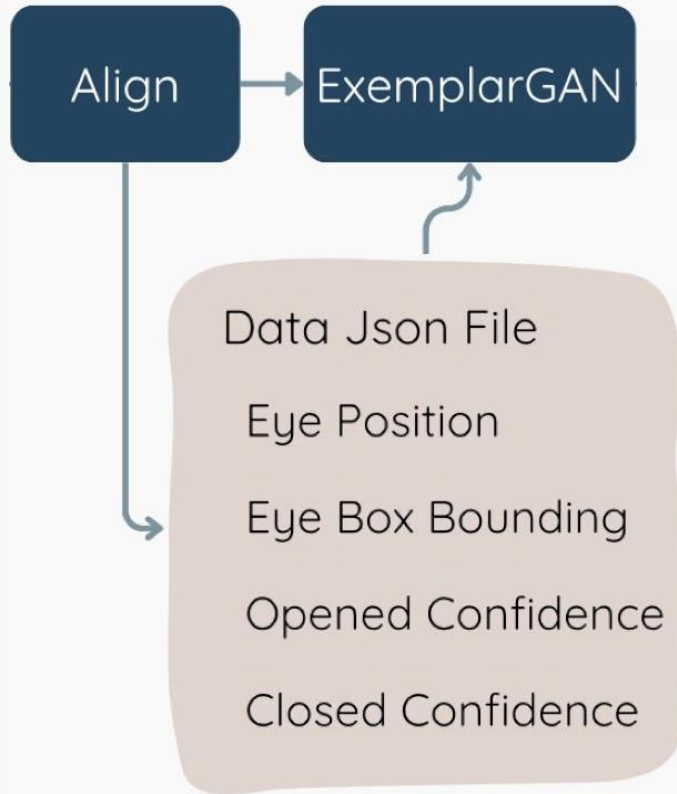


Main Approach – Cropped Face



- Factor
 - Faces must be clearly visible
 - No overlapping of faces between individuals is allowed
- Input Two Image
 - Identity Image
 - Reference Image
- Using OpenCV's DNN module with the Caffe framework to detect and crop faces and writing a JSON file containing the top-left starting point and dimensions (width and height) of each cropped face.
- Using Python and the Face Recognition library to compare faces and add labels.
- Since the cropped images may have low resolution, apply super-resolution to enhance their quality.

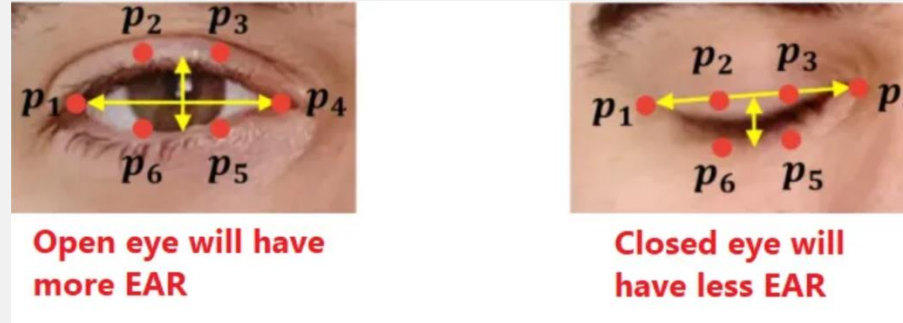
Main Approach – Align



- Using Dlib to Detect Eyes
- Record Right Eye and Left Eye Positions
- Calculate EAR Value
- Calculate Confidence Value
- Save as JSON File
 - Left Eye Position
 - Left Eye Box Bounding
 - Right Eye Position
 - Right Eye Box Bounding
 - Opened Confidence
 - Closed Confidence

Main Approach – Align (Cont.)

Calculate EAR Value

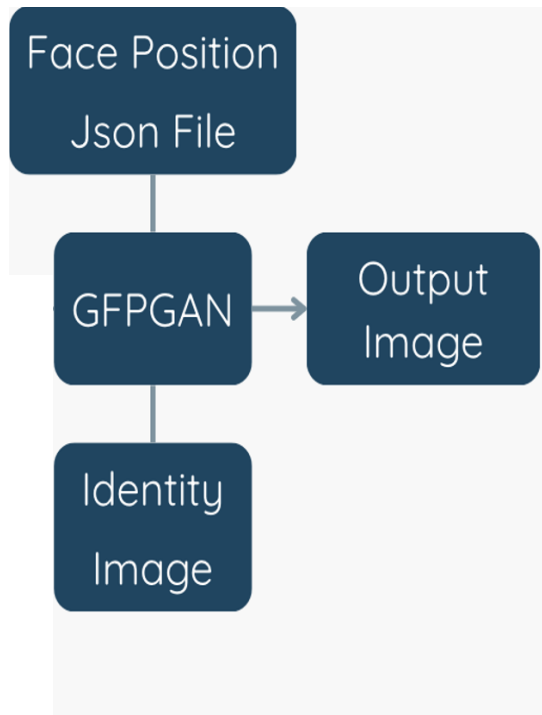


$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

Calculate Confidence Value

$$C_{opened}, C_{closed} = \begin{cases} (1.0, 0.0) & \text{if } EAR \geq T_{opened} \\ (0.0, 1.0) & \text{if } EAR \leq T_{closed} \\ \left(\frac{EAR - T_{closed}}{T_{opened} - T_{closed}}, 1.0 - C_{opened} \right) & \text{if } T_{closed} < EAR < T_{opened} \end{cases}$$

Main Approach – Map



Face Position
"face_index": int
"startX": int
"startY": int
"endX": int
"endY": int
"width": int
"height": int

- Based on the Face Position JSON file cropped from the identity image, paste the results back accordingly.
- Apply Gaussian Blur, Match Color, and seamlessClone to the pasted images to ensure they blend naturally and avoid looking out of place.

Evaluation Metric

Qualitative

- Naturalness
 - Check whether the generated eyes maintain harmony and realism with the original face
- Consistency
 - if the pasted effect has visible borders

Quantitative

$$PSNR = 10 \cdot \log_{10}(\frac{MAX^2}{MSE})$$

$$MSE = \frac{1}{m \cdot n} \sum_{i=1}^m \sum_{j=1}^n [I_1(i, j) - I_2(i, j)]^2$$

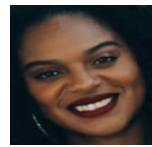
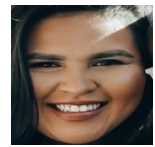
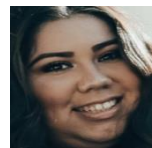
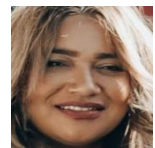
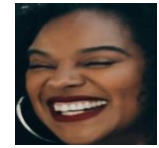
PSNR measures the quality of reconstructed images.

Higher PSNR indicates better image fidelity relative to the reference

Result and Analysis



Face Position



Identity Image

Cropped Face Identity Image

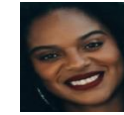
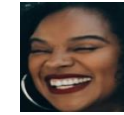
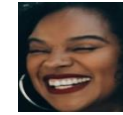
Face Recognition

Cropped Face Identity Image

GFPGAN

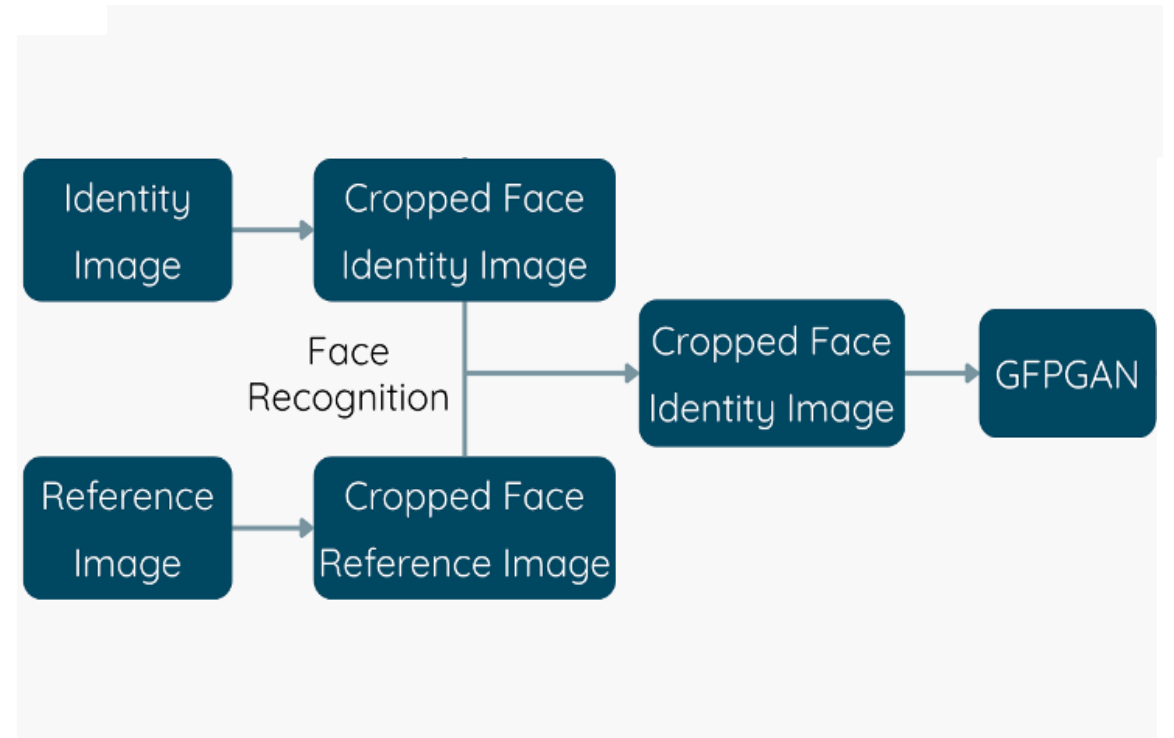
Reference Image

Cropped Face Reference Image

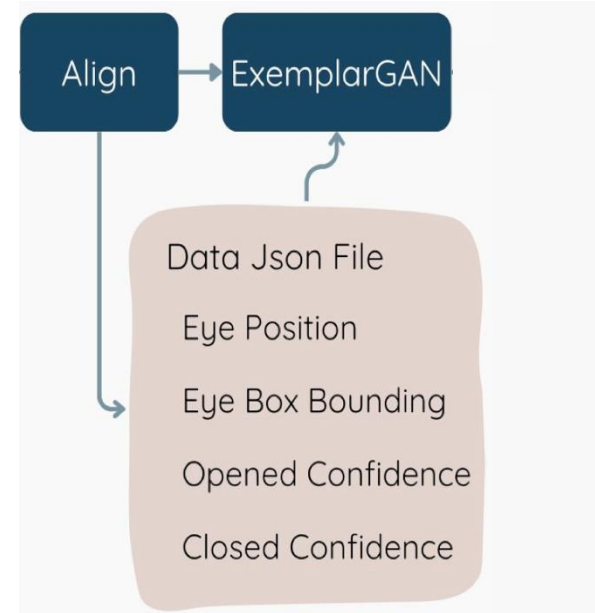
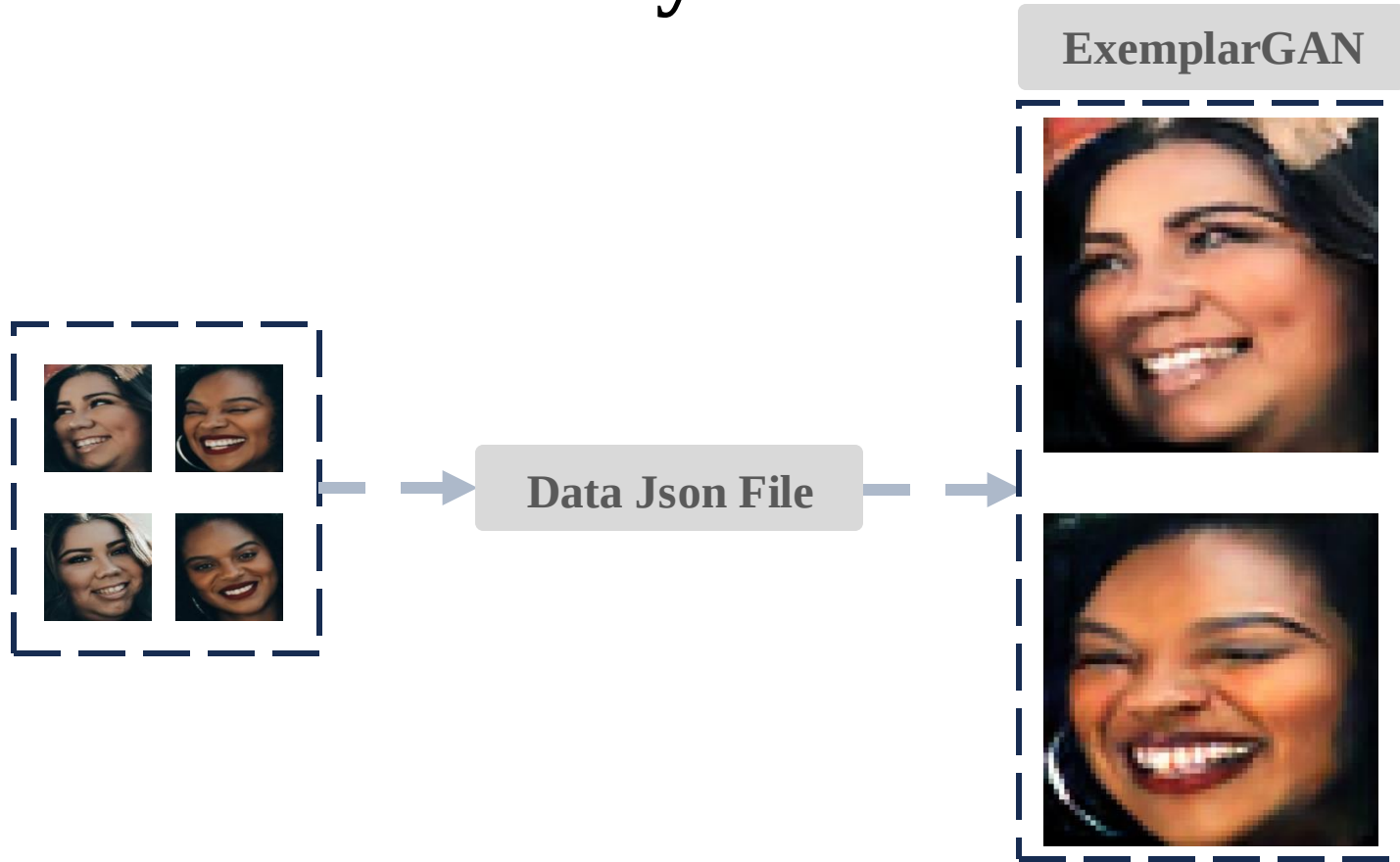


Result and Analysis

The number of faces cropped from the identity image and the reference image is different. This discrepancy occurs because some faces in the identity image are significantly obscured, making it impossible to detect them.



Result and Analysis



Result and Analysis



Face Position



Face Position
Json File

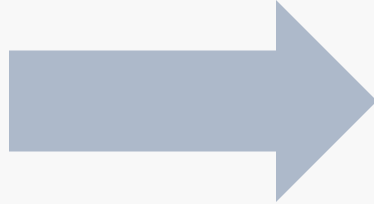
GFPGAN

Output
Image

Identity
Image

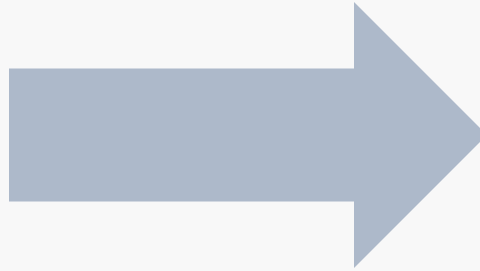


Result and Analysis



The repaired eyes show significant improvement compared to the original closed-eye photo, appearing slightly open, though the expected outcome is to have the eyes fully open. However, since the baseline model does not provide a pre-trained model and the training time for each epoch is relatively long with large datasets, this model was only trained for 220 iterations to achieve the current results. With further training, it is expected to achieve even better outcomes.

Result and Analysis



PSNR: 38.60 dB



Result and Analysis

Identity

Reference

Result



Identity

Reference

Result



Identity

Reference

Result



Identity

Reference

Result



Result and Analysis

Identity



Reference



Result



PSNR: 25.11 dB

Conclusion

Compared to the original image, our generated results show minimal differences in regions other than the eyes. From a qualitative perspective, the generated images are highly consistent with the original ones. From a quantitative perspective, the PSNR values of the generated images are all above 25, indicating a very small difference from the original images and excellent overall blending performance. This demonstrates that our main approach has achieved remarkable success.

The only shortcoming at present lies in the eye restoration, which has not yet fully met our expectations. Nevertheless, even with a limited number of training iterations, the model has achieved slight eye-opening effects, showing that it is improving in the right direction. We will continue to train the model with the aim of achieving the desired restoration results.

Contribution

江仲恩 (313553044) 50%

黃瑜明 (313553037) 50%

Reference

- [\[1712.03999\] Eye In-Painting with Exemplar Generative Adversarial Networks](#)
- [\[2101.04061\] Towards Real-World Blind Face Restoration with Generative Facial Prior](#)
- <https://unsplash.com/photos/2-women-sitting-on-black-chair-KodMXENNaas>

Tensorflow ExemplarGAN Github LINK

- <https://github.com/zhangqianhui/Exemplar-GAN-Eye-Inpainting-Tensorflow.git>